$\mathbb{R}^{2N_c+1}$, the W_i are independent Wiener processes, and D is the "deterministic part:"

$$D = \begin{bmatrix} \xi \boldsymbol{p} \\ -\boldsymbol{p} \\ -\boldsymbol{p} \cdot \nabla U - \xi \end{bmatrix}.$$

Then we calculate directly

$$D' = [D, Y_{N_c+1}] = \begin{bmatrix} \boldsymbol{p} \\ 0 \\ 1 \end{bmatrix},$$

and from this we see that $[D', Y_i] = \boldsymbol{e}_i$ for $i = 1, 2, \ldots, N_c$. Thus all the basis vectors of $\mathbb{R}^{2N_c+1}$ are found in the Lie algebra of the components of the Hörmander decomposition; as we know, this is the key step in the proof of ergodicity. Thus the method (8.33)–(8.35) is an alternative for molecular sampling dynamics, even though it is not based on a Newtonian model. Leaving the stochastic term off the final equation (i.e. in ξ) is easily shown to destroy the ergodicity property (the Hörmander condition is then violated).

Now consider the splitting into deterministic and OU parts. The deterministic part

$$\dot{\boldsymbol{q}} = \xi M^{-1}\boldsymbol{p}, \quad \dot{\boldsymbol{p}} = 0, \quad \dot{\xi} = -(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q})/\mu, \tag{8.36}$$

preserves volume and the function $U(\boldsymbol{q}) + \mu\xi^2/2$ and hence also the distribution with density $\tilde{\rho}_A(\boldsymbol{q}, \xi) = e^{-\beta(U(\boldsymbol{q})+\mu\xi^2/2)}$. By the same token the OU processes collectively preserve $\tilde{\rho}_B(\boldsymbol{p}, \xi) = e^{-\beta(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2+\mu\xi^2/2)}$. In combination the full dynamics preserves the extended canonical measure. If we iterate the two parts (assuming the deterministic part is solved exactly) the distribution will converge to the exact canonical distribution, due to especially to incorporation of $N_c + 1$ noise processes.

The OU process defining the fluctuations in $\boldsymbol{p}$ is completely decoupled from $\boldsymbol{q}$ and ξ, so it is perhaps most natural to think of the iteration as a random walk on the reduced space $(\boldsymbol{q}, \xi)$ where the directions are generated by an auxiliary OU process. Then the algorithm proceeds as follows: (1) choose a random direction $\boldsymbol{p}$ using the OU process, and (2) sample the reduced (stochastic) dynamics for $(\boldsymbol{q}, \xi)$ in the direction $\boldsymbol{p}$. We refer to this procedure as *stochastic line sampling* (SLS), for the following reason. With $\boldsymbol{p}$ fixed in step (2), the motion of $\boldsymbol{q}$ is confined to the straight line: $\boldsymbol{q} = \boldsymbol{q}_0 + \eta\boldsymbol{M}^{-1}\boldsymbol{p}$, where η is a scalar function of time. Inserting this coordinate change in (8.36) we see that the equations reduce to $\dot{\eta} = \xi$, $\dot{\xi} = -\partial/\partial\eta U(\boldsymbol{q}_0 + \eta\boldsymbol{M}^{-1}\boldsymbol{p})$. This is a one-degree of freedom system and hence is theoretically exactly solvable in quadratures, although the complexity of the force will render this impractical in calculations, and, as mentioned above, an additional stochastic process must be incorporated to drive ξ.

The deterministic part may however be approximated in various ways. We may solve it by splitting into two (volume preserving) parts, integrating first $\dot{\boldsymbol{q}} = \xi \boldsymbol{M}^{-1}\boldsymbol{p}$, where ξ and p are fixed (thus a drift along a given direction), followed by solving $\dot{\xi} = -(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U/\mu$, which is a similar drift in the auxiliary variable. Composing these drifts successively in a leapfrog approach, we arrive at the deterministic sequence:

$$\begin{aligned}
\xi_{n+1/2} &= \xi_n - (h/2)(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q}_n)/\mu, \\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\xi_{n+1/2}\boldsymbol{M}^{-1}\boldsymbol{p}, \\
\xi_{n+1} &= \xi_{n+1/2} - (h/2)(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q}_{n+1})/\mu.
\end{aligned}$$

Coupling this to the stochastic evolution computed from solving the OU parts in both ξ and $\boldsymbol{p}$ yields an alternative to Langevin dynamics that is 2nd order accurate in its steady state distribution and long term averages. Let us refer to this as the stochastic line sampling with leapfrog method.

Going a step further we may again obtain methods which exactly preserve the canonical measure based on canonical coordinate splitting, resulting in systems of the form

$$\dot{q}_i = \xi m_i^{-1} p_i, \qquad \dot{\xi} = -p_i U_{q_i}/(\mu m_i).$$

Up to time-rescaling by p_i, these are in fact the same types of systems as we encountered previously in our treatment of J-splitting of the Hamiltonian system. If desired, an rth order area-preserving approximation of the q_i, p_i dynamics could be substituted. This method results in paths that differ slightly from the linear path in the direction $\boldsymbol{M}^{-1}\boldsymbol{p}$. In simulations we found that this method, which we term "stochastic line sampling with coordinate splitting," was more stable than the leapfrog version when implemented using a further leapfrog splitting in each coordinate plane. However, we emphasize that both methods are highly experimental and are presented here as illustrations of the flexibility available in constructing methods to sample the canonical distribution—they have not yet been shown to be beneficial for molecular sampling applications.

We compare the trajectories and invariant distributions of the non-Newtonian line sampling thermostat (8.33)–(8.35) with three Newtonian-based thermostats: Langevin dynamics (7.4), Nosé-Hoover (8.2)–(8.4) and Nosé-Hoover-Langevin (8.27)–(8.29) dynamics. We seek to canonically sample a one-dimensional system with double-well potential energy function given as

$$U(q) = (q^2 - 1)^2.$$

Choosing all arbitrary system parameters to be one (i.e. setting $\beta = M = \gamma = \gamma' = \mu = 1$ for the methods), we select initial conditions $q(0) = p(0) = 1$ and $\xi(0) = 0.5$ with a timestep parameter $\delta t = 0.01$ sufficiently small to eliminate the possibility of numerical errors warping the results.